



GumBall6900

The index fund
that chooses itself

An immutable signal-directed onchain fund with continuously clearing GBX mining, tenure-locked miner rates, oracleless acquisitions, and in-kind redemption.

20M

GENESIS GBX

1-16

MINING SLOTS

80 / 20

HANDOFF / RESONANCE

Fixed

TENURE RATE

READ THIS FIRST

Development design, not a deployed system.

GumBall6900 is experimental software. It is not deployed, has not received an independent external audit, and is not authorized for user funds. Exact Mine economics, target-chain dependencies, legal provenance, and a signed deployment manifest remain unresolved.

This edition describes the architecture introduced by ADRs 0031, 0034, 0035, 0036, and 0037, whose core Solidity is implemented and covered by the current suites. ADR 0036 supersedes ADR 0032's fixed-rate rule. External governance remains unselected. A local green build is engineering evidence, never a safety, audit, or release claim.

EDITION

v0.7, 21 August 2026. Not deployed and not authorized for user funds. Independent external audit not completed.

Genesis issuance	20,000,000 GBX for canonical liquidity
Later issuance	One permanently bound immutable Mine
Supply limit	No economic cap; GBX has no voting checkpoints
Governance	External Resonance owner unselected; sGBX IVotes retained
Legal status	Farplace, give.fun, Liquid Signal, and transitive lineage remain unresolved release blockers

ORIENTATION

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PART I

01 The economic loop

Mining brings USDG in; signal-backed sGBX coordinates acquisition weights and exposes IVotes checkpoints.

- 01 A participant takes a mining slot at its current hourly decaying USDG price.
- 02 The incumbent accrues GBX continuously at the fixed rate assigned on entry.
- 03 A signal atomically escrows GBX, mints non-transferable voting sGBX, and assigns every unit to one live Strategy.
- 04 Twenty percent of a nonempty-slot handoff routes through Resonance; eighty percent becomes a displaced-miner claim.
- 05 ResonanceRouter waits below the active amount left; a qualifying balance restarts seven days with new USDG plus that remainder.
- 06 Strategies pull released USDG; each acquired-asset payment uses the current global 0%-to-20% Bribe rate and its Fund complement, with 1e36 reward-index precision.
- 07 A GBX holder may burn GBX for a selected pro-rata basket of raw Fund assets.

EMPTY SLOTS

A first occupation routes 100% of its USDG payment because no displaced miner exists.

NO TEAM FEE

Mining has no team, founder, management, or protocol fee.

PART II

02 A continuously clearing market

Each independently replaceable slot falls from its opening USDG price to zero in one hour.

A miner buys the right to accrue GBX until replacement and the possibility of receiving 80% of the next payment. The second value is uncertain: without a successor, no handoff claim arrives. This rollover risk disciplines the price miners are willing to pay.

After each handoff, the next opening price is the paid amount multiplied by an immutable factor, with an immutable minimum. Expected GBX value, market liquidity, gas, replacement probability, and contract risk all influence the real clearing price.

MOMENT	CONTRACT OUTCOME
Start of hour	Price equals the slot opening price
Thirty minutes	Approximately half remains
One hour or later	Price is zero; replacement still allowed
Nonempty replacement	80% claim / 20% Resonance

NOT GUARANTEED

Profitability, a successor payment, GBX liquidity, and frequent replacement are market outcomes—not contract guarantees.

CALLER BOUNDS

Every purchase supplies an expected epoch, a deadline, and a maximum price, so a miner front-run at purchase cannot be charged the reset opening price.

PART III

03 Miners keep the rate they bought

A slot rate changes only when that slot changes hands.

Claims, cumulative-mining thresholds, redemptions, and other slots' handoffs never rewrite an occupied slot's GBX-per-second rate.

STATE	EARLIER INCUMBENT	NEW TENURE
Global 100 GBX/hour	6.25 GBX/hour	6.25 GBX/hour
Global rate halves	Still 6.25 GBX/hour	3.125 GBX/hour
Incumbent is replaced	Tenure ends	3.125 GBX/hour

All sixteen slots divide the current global rate by sixteen when a tenure begins. A threshold crossing changes only the rate offered at a later handoff, never an incumbent's already-assigned rate.

ACCEPTED TRADEOFF

Aggregate issuance can temporarily exceed the current global rate while old high-rate tenures remain.

PART IV

04 Accrued mining counts before redemption

Fund reads Mine's constant-time effective supply before taking the common snapshot.

GBX begins with 20,000,000 genesis tokens. Its only later issuer is the permanently bound Mine. Global rates offered to future occupants halve at immutable cumulative-mining thresholds, then continue at a positive tail on every modeled horizon. GBX supports permit approvals but has no governance checkpoints; votes begin only after signaling into sGBX.

Rewards accrue continuously but each slot mints only when it changes hands. Fund reads Mine's constant-time effective supply before every redemption, so already-earned unminted GBX cannot be omitted from the denominator and redemption never depends on touching all slots.

PAYOUT

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floor(Fund token balance × GBX burned ÷ effective  
pre-burn supply)
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CONSTANT-TIME SUPPLY

Mine maintains one aggregate TPS accumulator across 16 fixed slots; Fund performs no slot loop or mining mutation.

OMISSIONS

Assets a redeemer leaves out remain permanently for the post-redemption supply.

PART V

05 Immutable by design

The core exposes four Resonance administration methods; external execution rules remain unselected.

RESONANCE OWNER ACTIONS

- Add a Strategy
- Permanently kill a Strategy
- Register a Bribe reward token, up to eight
- Set the global prospective Bribe share, 0%-20%

ABSENT POWERS

- No Mine administration or incumbent repricing
- No emission setter or replacement authority
- No core Governor, Timelock, or generic executor
- No Fund withdrawal or liquidity NFT recovery

SignalGBX retains block-clock ERC20Votes checkpoints, including historical weight after signal exits, but this repository does not select or implement the governance system that will consume them. A later ADR must pin the exact external executor, permissions, voting rules, upgrades, delay, cancellation, and ownership handoff. After bootstrap, the final live Strategy cannot be killed until a replacement is added. A rate change affects only later payment classifications; 0% disables no signaling or independently funded reward path.

RELEASE STATUS

External governance selection, independent audit, parameter review, provenance clearance, and signed deployment evidence are still required.